

Splunk Journey – Day 01

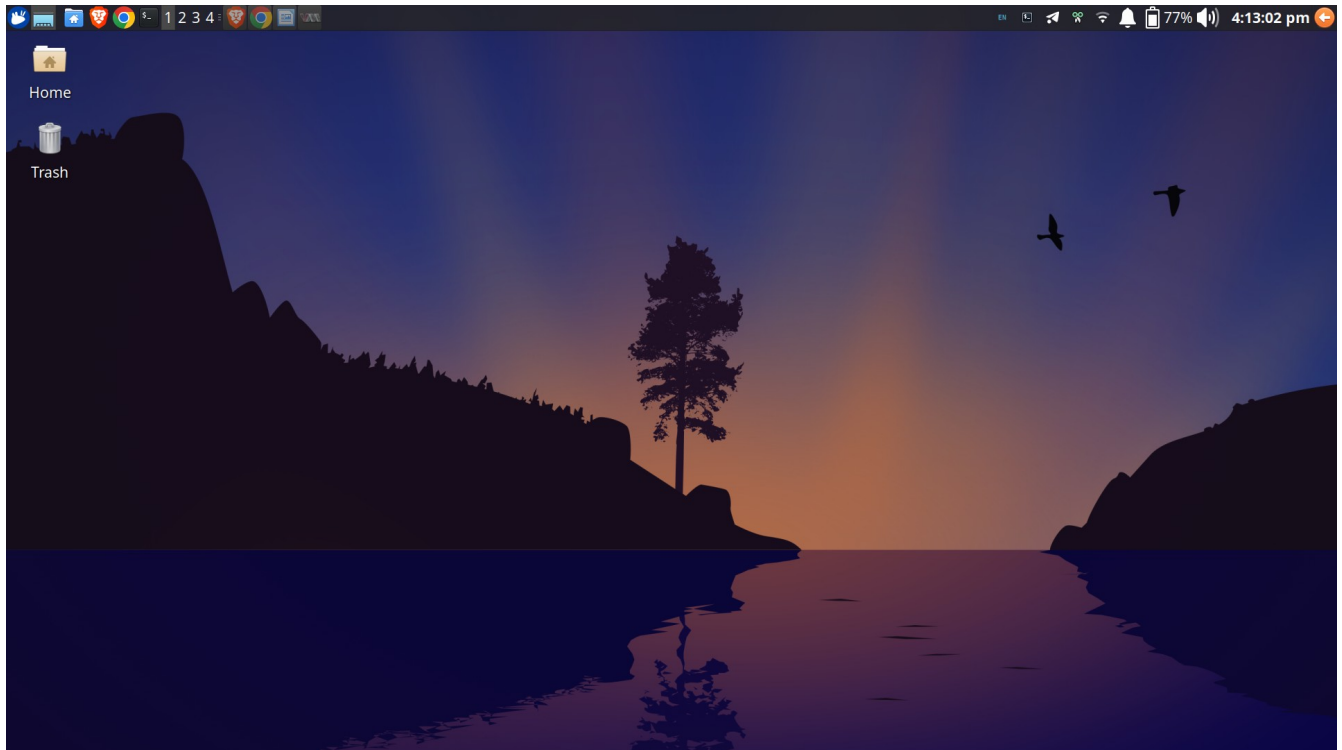
Date: 2026-06-10

Home Lab Setup and Splunk Installation

Lab Environment

Host Machine

- OS: **Xubuntu**
- Hypervisor: **QEMU/KVM**



```
status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: load (/usr/lib/systemd/systemd) (enabled; preset: enabled)
   Active: active (running) since 05:39:36 EDT; 1min 17s ago
   Invocator: Doc
   Process: 730 (sshd)
   Main PID: 1
   Tasks: 1 (limit: 4630)
   Memory: 2.6M (peak: 3.5M)
   CPU: 60ms
   CGroup: /sys/fs/cgroup/systemd/systemd[1]
           └─730
Jun 10 05:39:36 systemd[1]: Started OpenBSD Secure Shell server...
Jun 10 05:39:36 sshd[730]:
Jun 10 05:39:36 sshd[730]:
Jun 10 05:39:36 systemd[1]: Started ssh.service - openbsd secure shell server.
```

Virtual Machine

- OS: **Debian Minimal**
- Access Method: **SSH** from **Xubuntu Host**

```
File Edit View Terminal Tabs Help
ssh
The authenticity of host
This host key is known by the following other names/addresses:
 ~/.ssh/known_hosts:1
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added
 to the list of known hosts.
@
Linux
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

Objective

Install **Splunk Enterprise** on a Debian Minimal VM and verify that it runs correctly under the dedicated **Splunk** service account.

Prerequisites

Create a Splunk Account

Create a Splunk account to access **Splunk Enterprise** downloads.

Download Splunk Enterprise

Download the appropriate Debian (.deb) package and transfer it to the Debian VM.

Example:

```
wget <splunk-download-url>
```

Verify the downloaded package:

```
ls -lh splunk-*.deb
```

```
~$ ls -lh splunk-*.deb
-rw-r--r-- 1 root root 1.3G May 18 13:57 splunk-*.deb
~$
```

Step 1: Install Splunk

```
sudo apt install ./splunk-*.deb
```

Step 2: Verify Installation

Check whether Splunk was installed successfully.

```
ls /opt/splunk
```

```
ls /opt/splunk
bin      lib      README-splunk.txt  var
copyright.txt  license-eula.txt  share
etc       openssl  splunk-
include   opt      swidtag
```

Verify the Splunk user account.

```
grep splunk /etc/passwd
```

```
~$ grep splunk /etc/passwd
splunk:x:1000:1000::/opt/splunk:/bin/bash
~$
```

Check directory permissions.

```
ls -ld /opt/splunk
```

```
:~$ ls -ld /opt/splunk
-rwxr-xr-x splunk splunk 4096 Jun  9 17:09 /opt/splunk
:~$
```

Step 3: Fix Ownership

Assign the installation directory to the Splunk user.

```
sudo chown -R splunk:splunk /opt/splunk
```

Step 4: Test Permissions

Verify that the Splunk user can write inside the installation directory.

```
sudo -u splunk touch /opt/splunk/var/testfile
```

```
:~$ sudo chown -R splunk:splunk /opt/splunk
[sudo] password for :
:~$ sudo chown -R splunk:splunk /opt/splunk
:~$
```

Step 5: First Startup

Start Splunk for the first time and accept the license.

```
sudo -u splunk /opt/splunk/bin/splunk start --accept-license
```

During this step:

- Splunk initializes its directories
- Admin account is configured
- Web interface becomes available

Step 6: Basic Service Management

Stop Splunk

```
sudo -u splunk /opt/splunk/bin/splunk stop
```

```
:~$ sudo -u splunk /opt/splunk/bin/splunk stop
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
...
Stopping splunk helpers...
Done.
:~$
```

Start Splunk

```
sudo -u splunk /opt/splunk/bin/splunk start
```

```
:~$ sudo -u splunk /opt/splunk/bin/splunk start
Splunk> The IT Search Engine.
Checking prerequisites...
  Checking http port
  Checking mgmt port
  Checking appserver
  Checking kvstore p
  Checking configura
New certs have been genera
  Checking critical directories... Done
  Checking indexes...
    Validated:
the fishbucket history main _internal _introspection _metrics _metrics_rollup _telemetry _
Done
  Checking filesystem compatibility... Done
  Checking conf files for problems... Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunk/splunk-
  All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Done
Waiting for web server at http://127.0.0.1:8000 to be available..... Done
If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com
The Splunk web interface is at http:,      8000
```

Check Status

```
sudo -u splunk /opt/splunk/bin/splunk status
```

```
:~$ sudo -u splunk /opt/splunk/bin/splunk status
splunkd is running (PID
splunk helpers are running (PIDs: 928 1137 1142 1217 1265 1536).
:~$
```

Step 7: Create a Shortcut

Bash Alias

```
echo 'alias splunk="/opt/splunk/bin/splunk"' >> ~/.bashrc
source ~/.bashrc
```

System-wide Symlink

```
sudo ln -s /opt/splunk/bin/splunk /usr/local/bin/splunk
```

```
:~$ echo 'alias splunk="/opt/splunk/bin/splunk"' >> ~/.bashrc
source ~/.bashrc
:~$ sudo ln -s /opt/splunk/bin/splunk /usr/local/bin/splunk
ln: failed to create symbolic link '/usr/local/bin/splunk': File exists
:~$
```

Step 8: Manage Splunk Using the Shortcut

Start:

```
sudo -u splunk splunk start
```

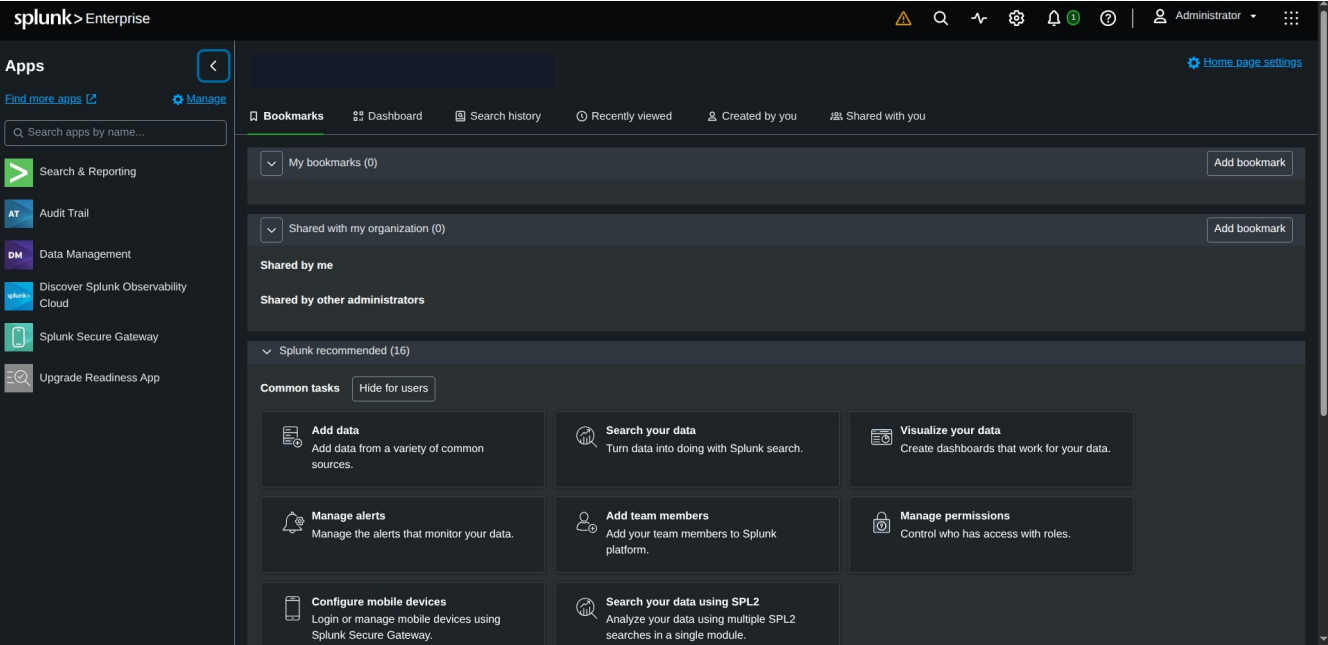
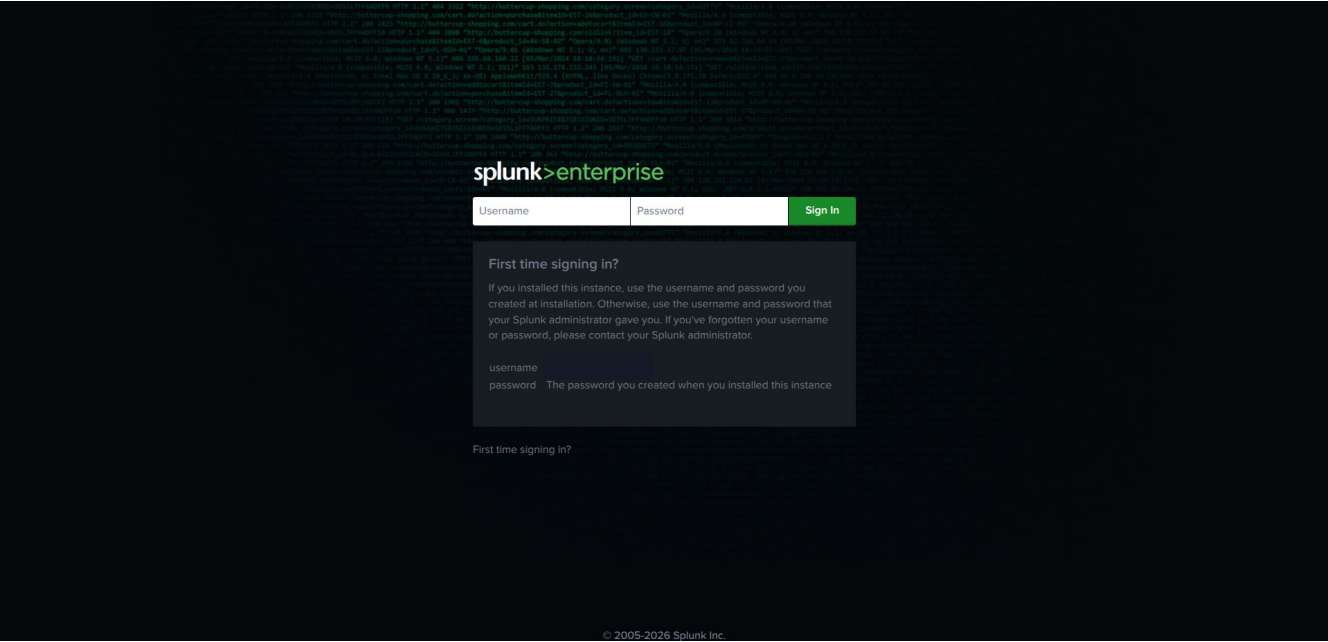
Stop:

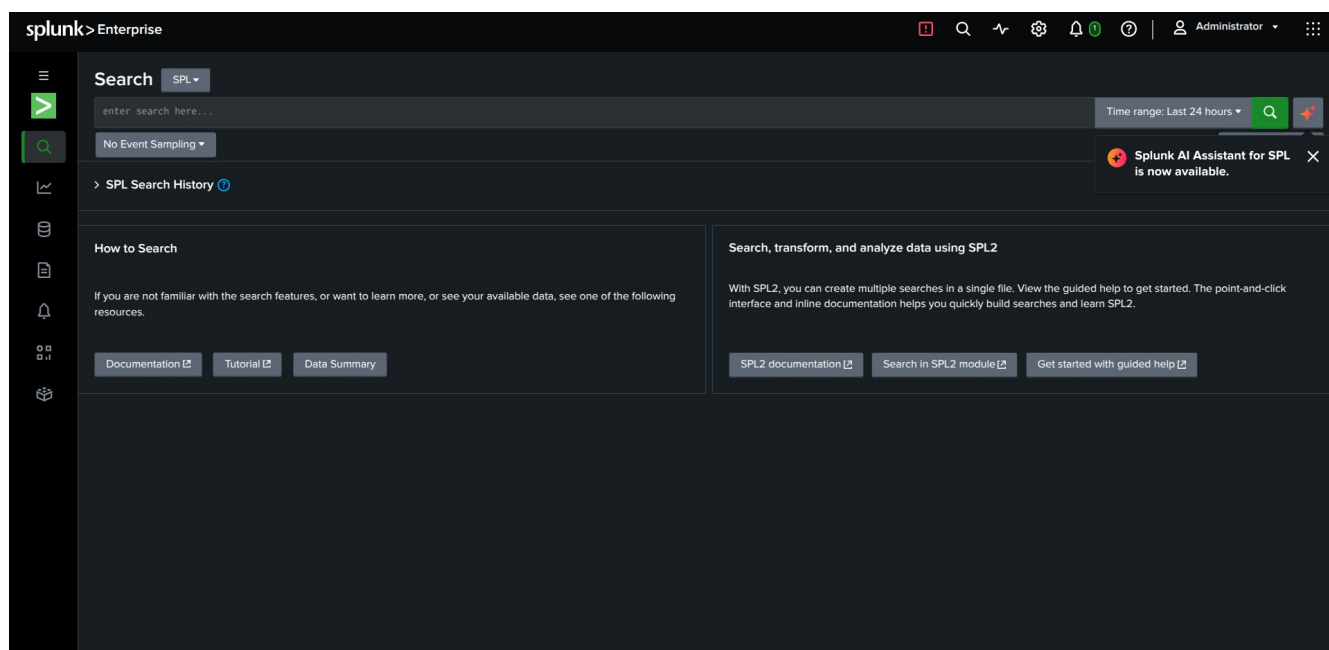
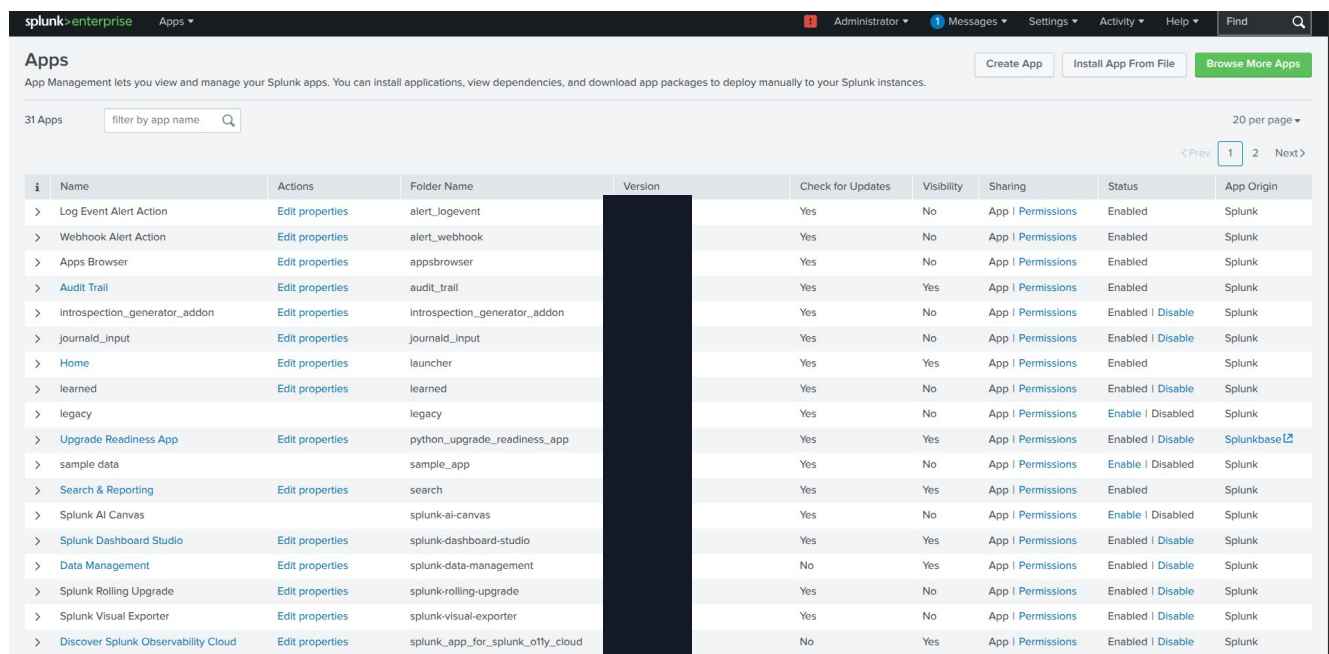
```
sudo -u splunk splunk stop
```

Check status:

```
sudo -u splunk splunk status
```

Interface





Optional

Auto-start Splunk when the system boots:

```
sudo /opt/splunk/bin/splunk enable boot-start -user splunk
```